

# MJ\_Interview 3.wh\_shambibble

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## SUMMARY KEYWORDS

prompt, talking, image, bit, problem, journey, ai, good, figured, words, mid, discord, idea, pretty, dolly, people, natural language processing, text, thinking, point

## SPEAKERS

Watson Hartsoe

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 Watson Hartsoe 00:05

Progress Live turns, enable live transcription. Can you can talk a little bit to see if it works.

 00:17

I don't know what I'm supposed to say I suppose for sounds very bad or it gets transcribed very bad. You'll have figured it out by now.

 Watson Hartsoe 00:23

Oh, it's it's working AI at its finest. You know, it's like it's like all happening at the same time. So this is a Yeah, it's so much easier to have it transcribe it for you than to sit there for hours. But um, yeah, I guess again, thanks for making time. I get I really got got high recommendations from God at play Joshua. So yeah, he read, he pointed out this book that you're writing, and I got to look through it. And honestly, I can say, I've already learned a lot.

 00:53

Really, it's more like, it's what happened. The thing is, is there we had a, we've been kind of like, chucking things into various ad hoc places. Like, there's an FAQ thread and FAQ thread became a forum ProvoCraft forum. But there's still like a, because the official documents, they explain things like in a bare minimum of but they don't really, like give examples, or, you know, which is fair, like they busy actually making the damn thing. So people were just asking, like, is there a place that just like, you know, explains how this works, and a more because people would like, come to with specific problems. Like, you know, what, why aren't Why aren't I getting a photorealistic picture? When I put photorealistic in it enough people ask that I figure Okay, well, I should put that into an FAQ. And then I just started kind of a wrote the image prompt thing. Chapter at first. That's been around a while. But eventually, people just were like, Well, is there a place where it's kind of just all is, and I was like, Okay, well, I'll try.



Watson Hartsoe 01:53

So that's what I did. Nice. Is it? Are you? What's your background? And how did you? I mean, like, Are you a writer?



02:02

I'm a patent attorney. Oh, no. Wow,



Watson Hartsoe 02:05

you have some insight, then into copyrights for for? Well,



02:10

the thing is, what the thing I have insight into is this copyrights or like, I don't, I mean, maybe I have some insight, but a lot of it is unsettled. And I can't, how it gets disposed off is going to ultimately be a political decision, probably. But what I think it were, I think it has helped is the facility with describing a thing. Using abstract language in a way that's very likely to be misinterpreted. If it ever goes to court. This is a skill that did not have much overlap with art until about six months ago. Wow.



Watson Hartsoe 02:48

Okay, this is this is really insightful. So basically, like your job is to technically describe, you know, images or systems. And now that's given you this creative superpower. In some regards.



03:01

Nothing is created, it's more just like, like, thinking really hard about, you know, because most people they use words, they don't think really hard about double meanings or ways they can be misinterpreted or things like that. So, like one of the, one of the tougher things to do on mid journey, if you're just starting out and are just like trying to, it's ultimately an English problem is, let's say you want the color orange, but you don't want an orange. That's the thing you have to figure out. Because if you just put orange, you



03:34

will get a fruit.





03:37

So there's different ways you can do that. You can sure you could just like negate fruit in your prompt no fruit, you might have to stack it up a little bit, because it's a very strong association. So be like no fruit, no sphere, no circle, that kind of thing. That doesn't really, it doesn't really apply very much to how patents or patent claims are drafted. Except insofar as you're thinking adversarial, you're thinking, Okay, well, how is this gonna be screwed up?



Watson Hartsoe 04:01

Hmm. Interesting. So the adversary, in some ways is like, all of the associations that are in the training set. That could be I don't know, it's like the token could be parsed in a different way than what you're intending it to be? I don't know if that's right. Like I think, okay, okay. Yeah. I mean, like, I appreciate it, just understanding like, reading through your, your manual, like just like the idea of tokens existing independently of words, and sort of like, yeah,



04:29

because stable, the fusion has had that for a while now, where if you put in a prompt, you can, it will, it will spit back the tokenized version of that prompt, it will show you which words got run together and which words got split. And I assumed they could do that here. I don't know if that's just something they haven't thought about, or is it a priority? Or if they've made a conscious decision not to do it for I don't know, creative reasons or whatever. Maybe they refresh the tokenizer a little more often than they do the training. I don't know there's a bunch of reasons. But yeah, it's one thing I wish that they had here.



Watson Hartsoe 05:02

I want to I want to like, I want to dive deeper into this because I think this is really what I do want to talk about. But I also want to like ask a few just sort of like introductory questions about, like, for example, like how do you prefer for us to display your name and our paper? You want us to use your user ID your real name, a pseudonym, what's your preference?



05:21

Anyone who like if you pretty recognizable at this point, so cool, because I don't I don't have a public mode on I have private mode on. So when I, you know, make something public, it's generally pretty, like cool and stuff. So and I've been around the discord posting way too much for a long time. So like, people will just know, oh, this is a sham nipple thing.



Watson Hartsoe 05:43

Okay, so your user ID sham bubble is, is adequate. Okay. That's awesome. And I mean, like, these are just sort of like the opening questions.



05:53

Some of the things that they're I think I was the only one to first one to do the recurring face image prompting and the first one to do the word reinforcement. I think I could be wrong. Yeah.



Watson Hartsoe 06:05

I mean, like, how long have you been on the journey? Or when did you start using the journey? I started



06:09

it in like, I started my trial in like, early June, but I didn't really get into it until a couple of weeks later. How did you? Oh, sorry. Someone was someone was posting them in another discord and was handing out invites?



Watson Hartsoe 06:23

Do you remember what your first prompt was?



06:27

Yeah. Stick Figure



06:28

reenactment of the Battle of cannae? I think?



Watson Hartsoe 06:32

How did how did it turn out?



06:34

It turned out that well, there's there was a bit Did you see Did you see the YouTube video I made?



Watson Hartsoe 06:39

Oh, the the highlight reel I saw part of it. I think I remember some things.



06:43

There's a bit in there were like one of my first prompts was, like a ancient Egyptian depiction of a Miles Davis concert. And then they were holding a contest for free, free fast hours. So I was like, what, what what of my early first couple of prompts would be cool to revisit. So that's when I made the like badass, you know, ancient Egyptian depiction of a Miles Davis car was originally going for Sun raw. But again, as it turns out, there's a lot of other associations if you're trying to prompt the Jasmine Sun Ra, in an ancient Egyptian setting, you're going to have a very hard time not getting just like a guy with a bird head and a big glowing sun. So I eventually just gave up on that idea and went back to Miles Davis.



**Watson Hartsoe** 07:21

I've definitely used the sunrise style before. It's a pretty I think it's pretty strong. Like it definitely captured something



07:30

strong instead, that was just why it's not good. If you want a you know, an obscure jazz musician named son raw.



**Watson Hartsoe** 07:35

That's true. That's true. I mean, I'm curious, how did you learn to use in the journey? Or like, how would you describe your uses usage changing over time?



07:43

I just, I don't know, I just mess around and figure stuff out. Like I, I was like, literally, just when I first read the manual about image prompting, I was like, Okay, well, this probably works like it does on other places. And so I put, you know, stuff in there. And it didn't really seem to get used so much. It was just like, it was more abstract until it was being used, it wasn't being like directly modified. So I can tell it were being abstracted a little bit. And then I got the idea What if I put my selfie in there? That didn't do much? Then I said, Okay, well, what if I put my selfie in there and prompted with a guy that seemed to do a little bit? And then I was just like, talking on Discord with some, some people and someone else piped up with Can I do a recurring character and something clicked in my head? And I was like, I probably could, because I just prompted like, my face and the prompt a guy. And it kind of looks like me, it's like a caricature of me, you know, it's not really me. It's not going to fool anyone. But you can tell it's inspired by it. So then I was like, Okay, well, let me take this other prompt that I have and put that face on this other thing, and boom, there you go. You have a recurring character, which is still, like I saw a New York Times article the other day, where they're talking about the future of AI and stuff. And one of the complaints they have as well, you know, you can't really do a recurring character on these things, unless you unless you prompt a celebrity or something like that. And I'm like, no, no, you can. It's kind of why I mean, it's, you know, I was, I don't want to give

people too hard a time but that did kind of annoy me when I saw those folks doing like, comics and mid journey and they're doing these interviews about how powerful my journey is not even turning down stylize as other characters, other girl characters saying. I've definitely



Watson Hartsoe 09:23

definitely had this complaint in office hours before where somebody was trying to I think it would be the character they landed on it was like this yellow character I remember them talking about like, it was like a yellow Elmo and like, I think they just were using this word that like consistently rendered something that looks similar, but how did you how did you stumble across this? I mean, I guess you just describe how you stumbled across it.



09:48

Like I literally I just try and like put things together in



09:52

like, ways to just ways to play with it.



09:56

I wasn't I wasn't, I definitely wasn't the first one to Do the, the the little hack that used to be a thing before remix mode was on that was that was told to me that was a thing but way before I learned about it, but I what I figured out what I figured out how to do specifically when they launched the test algorithm. And so this can't take image prompts, you can't like put a character in there anything like that. So I went, and I made a portrait with the character filled off the screen. And then I called up an ER, I found an old portrait, sorry, I didn't make one, I found an old portrait where a character took up on the screen, and I called up the job ID. And then I ran the job ID again, got the seed and the prompts, stuffed all those into a custom, which, again, is a thing you could do. But I was the one that pulled the old job ID. And then I remastered it for that was a thing, and then yelled about it a lot. So they made it a thing until they made it a button. So it's like mainly just like trying to figure out a way to I don't know, play with these things and hacking together and do things that, you know, maybe didn't even occur to the developers, which I'm sure it's like, I'm sure they would say that's part of the charm.



11:04

Yes, systems like these?



Watson Hartsoe 11:07

I mean, had you It sounded like you had been on Discord before before this, if this is how you heard about majority, but I guess what was your impression of this Discord server in particular,

when you first landed in it?



11:18

I mean, it's kind of big at this point. Like, it's one of those places where you can't really have an opinion on the server, because you can only ever really hang out in one or two channels. Otherwise, there's just too much there. Like I hardly ever, like a lot of the probably the bulk of the server is people. Because I put on private mode, it didn't really look back that part. But some people get more out of the social aspect. And so probably the bulk of the server activity is stuff I don't see which is people just like in the public channels, hidden remix and sometimes upscaled to their chagrin on others buttons, and playing and stealing bits of prompts and playing off ideas and stuff. I don't like I'm not like, precious about my stuff. Obviously, I share it and prompt crap and whatnot. I mainly have in a private mode. So like, it's just, like pictures of friends, friends. I'm talking to friends and family about the stuff. They're like, cool. Can you make me? And so I got private, most of the pictures wouldn't be public.



Watson Hartsoe 12:14

That makes sense. I mean, the prom craft channel, you've been involved in for a while, like, What is your impression of how prom crafting has



12:22

evolved? I don't know. It's I mean, there's part of the reason again, that I was trying to draft that official manual was there was a lot of people you know, there's a lot of people who come in and ask questions. And it's always it's a little bit bifurcated, because now there's a forum but they couldn't just delete the channel because it turns out that the forum, it's not ideal for like a chatting setting. So a lot of people will get referred to the forum if it's a specific recurring problem, but it's what it's it's fun, I think to just like, sit there and debug things. Like if someone wants to an hour an alligator in a human trench coat, you know, that's one of the that's, that's a specific problem. That's fun to play around with. And so you go and you mostly prompt like a private detective wearing a trench coat. Alligator head or something like that. Is that a thing? Was that



Watson Hartsoe 13:12

100? No, sorry, was it was a notification on my computer. Yeah.



13:16

Okay. Yeah. It's fun. It's fun to just like, you know, bounce things off and play off ideas. And sometimes I even use sometimes I even steal other people's characters. One of the original one of the quote unquote original characters in my roster of characters that I test props with is not original, because I i i stolen image prompting him from one of his incarnates prompts. But yeah, it's just, it's fun to it's fun to to debug people's problems and come up with cool stuff.

 **Watson Hartsoe** 13:45

I mean, yeah, like the, there's so many layers of ownership ideas, like the idea of the training set, the idea of like, the prompts the idea of the images that come from prompts, I guess, like, do you feel like well, I mean, like, definitely, if you're using the face of somebody that's, you know, a known entity, like that's something that is more private, but like, I guess how do you imagine the line between private and public

 14:08

with with language? What do you mean?

 **Watson Hartsoe** 14:12

Like, like, some people are super secretive about like prompts like special words or like ways of prompting or like other people share prompts,

 14:21

especially precious now, because these things are going to change so fast. Everything everything I'm talking about could easily just mean absolutely fucking nothing in six months. So you know it I don't think it's good to be too precious about these things. I don't know. I can't again, I I have maybe a little bit more of an informed decision, but or an informed opinion. But in this case, the informed opinion is studied ignorance because I really don't know how the IP is all going to shake out. No one really knows. I think that it's some of it is some of it is pretty clear, you know, precedent like you know, if you go and you Celebrities in your fan casting or whatever you try and make a comic out of that you're going to hear from their lawyers because publicity rights are a thing. It's the same reason you need permission to deep fakes someone in your in your car commercial. So that I think is going to be pretty straightforward. But things like the training set is just AI. Again, that's going to be completely up to whatever. Agency and judiciary ends up having having a say, because it there's there's very little, it's like that.

 **Watson Hartsoe** 15:34

Is that the FTC? I mean, I don't want to get too technical or speculative about this, because you're right. It's gonna be like,

 15:41

suddenly the Library of Congress has some say, are the ones who do copyright. But yeah, beyond that is, maybe it's maybe a Supreme Court case. And for years, who knows? Like, I think that the, the, the companies that have the most to lose in the short term are stock photo companies, artists, I think. I mean, I think there's there's mixed opinions on that. But I think

there's there's definitely a case for this as a tool as an incorporating it as a tool. And I don't think it's necessarily going to be as destructive as some fear. Companies will probably try that. But even as someone coming from someone who likes mid journey, you know, a lot, you can tell a lot of the output Hossaini, a lot of it needs a lot of love and post production. So, you know, I think it's a concept artists did best in the short term, but stock photo companies their their craft, if you just want like a vague illustration that doesn't need to be artistic or even interesting, beyond breaking up the text, which is what most stock photos are for. Yeah, someone like Shutterstock, or Getty is going to sue open AI, probably, since they're the ones that do photo realism the best. And also, because they're an entity with deep pockets, you can't really you're not going to get much out of Su and everyone who has a statement now. And yeah, they're gonna say, Well, you look, you used all of our shit in your training, there's even specific algorithms to wipe out recurring watermarks, that are openly talked about in your papers. You know, dollar sign gesture.

 **Watson Hartsoe 17:14**

Yeah, yeah, I mean, like, it seems like a pretty cut and dry, like the thing that is needing to be licensed, the output is competing against. So it's kind of seems pretty

 **17:26**

this is the other half of that though, because again, like I said, in the in the manual, these databases are not, they are not copies of the pictures, the copies of pictures are what they make available publicly online. So the argument from the other end is you can go online to Shutterstock, or Getty, or wherever. And you could call up one of these pictures, and you could look at it. If you're if you're learning how to draw yourself, you could use one of them even as a model because you know, you you mentally don't have a problem with painting the Shutterstock, watermark or whatever. And the all of these databases that they used to train them with do not actually have copies of the images they have. They have links to them, which are within a publicly available online. So the counter argument to that is like, Do you have a copyright in these images, but nothing is actually being copied, people are looking at what you have made publicly available. And so to extend the, you know, IP rights to training a database would be akin to extending copyright, not just to making a copy of something, but even to look at it. It starts trucking right into performance rights or something like that. So there's, like I said, I don't know how those two angles are gonna end up interacting. But it's not the I don't I really don't think it's kind of dry.

 **Watson Hartsoe 18:45**

I appreciate I appreciate your perspective, because the thing I just said to you is from like a, I guess, like a lecture from a Getty lawyer. So like, definitely makes sense that it was coming from somebody trying to defend one of these stock image. Everyone kind of make their but yeah, it's true. I mean, okay, so, so thinking about ownership thinking about copyright, that's definitely one side of this. Another sort of question I'm curious about is how you perceive the potential harm that these algorithms can create through other types of, I don't know, deep fakes or, or what have



19:21

you. Well, these algorithms in particular aren't much use for deep fakes. Because the real danger with deep fakes is video, because this is something that like we it's been true for a while now, for a couple of decades now, since Photoshop got good in the 90s that you can, if you care enough, you can Photoshop something, someone still image and that used to be a technical skill that used to be very difficult, but again, the Photoshop has gotten really good in the last 20 years since then. A lot of stuff gets can be done with AI a lot of stuff that used to be painstaking and you'd have to like, smudge out in individual, you know, stains in the 90s? Now you can just run a healing brush over it. So, hmm. But, you know, so that's been a thing for a while. And I think we've gotten used to it and people, you know, people already trying to use doctored photos and things like dirty politics and we're, you know, they've worked sometimes but their work on stupid people on who many people were kind of used to that we've gotten the nerd to that? Well, I don't know, the East particular engines have anything? Should have anyone scared the video I think is more troubling. Although I don't I don't know, I'm not in any AI. I'm not in any beta testing for those that can't get there yet. But it's probably only a matter of time. And video, you know, again, you can't you could fake it. But unlike Photoshop, it's still really hard to do, you pretty much got to be a professional effects guy to do a convincing, you know, to fake or at least you did nowadays. I know. Again, I don't know what the exact programs are. But I know there's things that people who don't, you know, we don't code who aren't like After Effects or Premiere pros or whatever. I'm used to like that on YouTube. They'll do like, you know, recap factory castings of scenes and stuff. So, yeah, that that I think will be that will be kind of a disaster because unlike, unlike Photoshop, the barrier is not going to glide down over several decades, it's going to be a pretty it's already a thing now. And you know, I think they saw the they tried to deep fake, I don't know, Solinsky snorting cocaine or something in the Ukraine. And it didn't really, you know, convince people because it's too on the nose. But one thing, no pun intended.



21:40

But, you know, the, yeah, I



21:43

think that's a lot more dangerous, but I can't really, I can't really speak to it. And I also don't know what to do, like, I don't know, like I was I was I was talking to a guy on Twitter about this. And it was like, you know, the stable diffusion was making waves because they, they sent it down to the ball. All releases are going to be safe for work, but their first release was heavily uncensored and had a ton of porn in the training data set.



22:09

So you could you were



22:11

getting, you know, tips in your face, even if you didn't prompt them. And if you if you want it to prompt some really bad stuff, you could. And obviously, this, you know, once again, one side of the coin is, well, this was irresponsible, you know, you you shouldn't have done that. But on the other hand, this stuff is only going to get cheaper and easier to make because the techniques are public, what's not, what's what's expensive. And restricting supply right now is the compute required to train these models, but the knowledge are public. And as you know, stuff as computers get faster and stuff gets better. It's gonna get easier and, you know, harder and harder to control. Because let's say that you are officious and sensorial congressman, or sorry, a snark. Sorry, sorry, it's congressman, who doesn't like that there's an AI out there that you could maybe prompt, you know, some horrible things are. And so we, you know, what are you going to do? Well, you have, you have basically only one choice, which is instituting People's Republic of China level, data controls, like a firewall, total ISP logging, and all that kind of stuff. Because, you know, it's, it's one thing to just do whack a mole, like you do in civil cases with copyrighted stuff. But if you're serious about stopping this thing from spreading, you would need like actual high level just surveillance state to stop it. And that's only going to become more true as these models get lighter and easier to process.



Watson Hartsoe 23:43

That's interesting.



23:44

I don't know what the I don't know what the control scheme is, I think that there's definitely going to be a huge societal downsides, for sure. minimize them. But I don't know what the alternative is. Like, you can say the alternative as well, you know, not betting them maybe, but just regulating them. But there's, again, you're gonna have that natural tension because you can only the more you regulate it, the more attractive the alternatives get. One of the things that was bugging me, is that they were talking about banning bikini pictures or something at one point. Now I'm not a guy who sits around and makes you know, a butt ton of Bikini pictures and that's all I do. But, and I can get around the filter Most times when it's not in you know, because they don't care if you try and get around the filter with synonyms and stuff as long as it's not intending to evade the rules. You're just continuing to evade. You know, you want to print off the blue footed movie or something, you know, the bird if you what you have to do to work around that, but that there's a ratchet that sort of only gets tighter. And so one of the things that was annoying me about the idea that they'll get Morrison Soria was that, you know, Dolly tune was already very frustrating to us, because in addition to the much stricter word filter, there's clear you can tell that it is heavily skewed towards towards cutesy stuff and against horror stuff, like if you prompt a Spider Woman, you know, on mid journey, you can get something that looks like outside of a Tim Burton movie Montale. A woman with like heavy mascara. And like a spider motif in the background or something, it's it's just, you can tell that it's kind of keeping people on rails. And I would like there to be middle grounds available, I would like there to be places where you could prompt you know, PG 13, or even R rated stuff without having to go to the same well of water as the people who just want a prompt like endless Gore, rate, Lolly porn, whatever. So I would like I think that's mostly good, I don't think you can control those things, I don't think you can stop those things. So I think most of the reinforcement is going to be social. And to have that social reinforcement, you have to give people some room to breathe. You have to have to be a little bit modest in your goals. And you only try to get cut off the worst of the worst stuff. Because anything is that if you start getting

broader, if you start getting puritanical, and you know, getting scandalized about bikinis, then it's, it's just gonna make it impossible to control. Or, you know, even in a societal reinforcement sense.



Watson Hartsoe 26:20

I mean, there's all types of even new cursed images, like things that you couldn't have even imagined being disturbing, but like, you're able to conjure up without even, you know, seeking out to conjure them sometimes. So like, I wonder how much it like not even playing Whack a Mole. It's kind of like, there's whole new species of mole that like, you know, might have like, weird, you know, maybe just



26:46

the other day on Twitter surfy. The novel AI hashtag, the novel AI is it's a, again, it's a stable diffusion fork. It's one one fairly loose rules. I haven't used it yet, because it's mainly animate to it. And I need a little bit more. I mean, I don't not make anime, but I need a little bit more variety, some kind of waiting on new journeys, animate model, because I like sometimes I like I'm very, I have a very broad taste. I'm not too enthused about their offer to train a style because I don't want to train a style, I want to have no style, you know. So but I was getting I was on Twitter, and I clicked the novel AI hashtag. And the first thing Twitter serves me is an account that's posting like little girls with their clothes getting torn off. And made on that day. I don't know if they actually made it on novel AI most likely they made it with because I doubt they allow that on their site, even though they are looser, most likely they made an oblique version of their mouth, which, again, you can't control this stuff. So yeah, there's just entirely there's going to be entirely because even though stable diffusion is said we're not going to do safe forward stuff anymore. Like I said, the knowledge is public datasets are out there. Someone you know, maybe right now only Emad can afford it. But in a year or so anyone you know, with for recommissioned levels of art despairs gonna be able to train their own AI. Hmm.



Watson Hartsoe 28:06

Yeah, this is it's curious. I mean, I think there's a lot I could talk about in terms of speculation. I'm also curious in like grounding this in some of your experiences with mid journey. Are there any particular biases or patterns? I know that you've talked about it in the manual? But I guess in your experience, where do you see those popping up most often?



28:25

Um, well, people,



28:28

most often it's people. It's, if you if you you can do things like one of my, like, if you prompt us, Somali woman, for instance, it will give her it will give her I don't know, pronouncing it right, but a black or whatever it is, that stuff? It will give you that even if you don't prompt it. Now, that's a

a hijab or whatever it is, that scarf? It will pursue that, even if you don't prompt it. Now, that's a bias. I don't know that it's that terrible bias. Because I'm sure it's a I mean, it's I think it's a Muslim country. So I'm sure a lot of women do wear that over there. But it's mainly just thinking about, you know, the AI is very naive. What stereotypes, you know, does the average person have because whatever stereotypes the average person has, is probably going to be the stereotype. And that journey has, because it doesn't curate its text description, period. So there's some decisions that they've made that have made it a little bit worse. I think, like I said, I think stylizes tuned a little too high. Because there's just a certain style of female portrait that's just like a young, pale skinned woman with black hair. It's just like that that's that was a type of thing that as type of style that was really popular with the early userbase of mid journey. And so when they incorporated the user ratings, it got really good and really cohere into drawing that face which means that you now see it on the default settings unless you turn the Stylize down. You will have a hard time I'm getting glasses on women's face at default Stein highs? Again, this is probably because of it's probably because of viewer ratings, not that I think like, unnecessarily the anti glasses bias is that strong these days, but the AI is that good at drawing? There's a lot of books, obviously, less of those are going to be rated higher. Some of the feedbacks are, you know, in a sense, relatively speaking, or at least not malicious. But they're still there. And, you know, I wouldn't, I would, I would prefer, they tune that down a little and tune the chaos up a little bit. That's a, ultimately, that's a personal preference. And I can't come down on them too hard for that, because I have bigger problems with their competitors. The bias in the data sets again, that that I agree, that's a problem. I think, right now, it's important to be vocal about the problems and to be conscious of it. But I don't know what you can do about it. You could, I mean, you could sit down and pay people, you know, to sit down and pay people a bunch of money to go through the database. But but the thing is, when you're talking about specs, stamping out bias, it's not all it's not, it's not as simple as just like killing a couple of keywords, or whatever you're at, because a lot of it again, a lot of everyday bias is subtle. And so I don't know how you score that. I don't know, even if you were doing it, again, encircling it with a bunch of Doc review, you know, temps, I don't know how you score something like that, or have a flowchart for it, you know, is this, is this reinforcing a stereotype? Well, you know, should you should you nuke it, all of us, you do get all you're gonna go the opposite way. So now you're in a position to Okay, well, what is the optimal ratio of, you know, demographic age associated with object to be, you know, you don't, you don't really know that you don't ultimately know the base rate, you're just you know, it's a little bit too high, you can you know, for at least for what you're going for, because you want your users to have a little bit more freedom. But, you know, it's tough to know when even if you're, if you could magic into existence, a huge workforce of humans how to do that. So I don't know that there's a good way to do that right now. Possibly, as the AI is get better. And the you know, the because recursion is already part of the training, AI evaluating AI and stuff, possibly is that consider it will take care of itself, I think it's definitely something people need to be conscious of. I just don't know that there is a knock that I saw with Dolly to Trump. And I had I don't have a problem with that. I think I suggested it at one point in the majority discord was if you're if you under specify a character, like a race, you know, specify a race? Well, it could be you do a little dice roll in it, and then you pass on a hidden prop with the ethnicity. So, you know, it's not always just generating a white guy, you know, when you ask for a CEO, or it's not always generating in their face when you ask for a terrorist or something like that. That has that that has its uses that also has its drawbacks. Maybe to give you an example, at one point, I was, again, I was just on a different discord. But I was bouncing ideas off of friend. And he wanted to see what the AI is that I had access to would do with his setting of like an alt history India, where India is really advanced and is more he's not headed for by European stuff. It's a cool setting. But he wanted to prop this whole history where India



33:23

was super advanced that had Gundams and stuff like that. So he asked me to prop was a see boy soldier. Or, you know, I don't know that this makes much sense because they worked for the British Empire and that but that's what he wanted, you know, so, okay, well prompt this prompt to see boy soldier holding a musket or something like that. And so I did that on mid journey and got a guy and an Indian guy and I did that on a table to fusion and got an Indian guy that I did it on dolly to and unfortunately, Dolly twos filters are not that smart. They don't know when something is they think something is under specified when it isn't and it didn't know what a CPI was sterically and so it gave me three white guys and then for variety sake, she got to have right it gave me a white girl that was probably gonna see boy soldier in a Victorian British military uniform. So sometimes that can sometimes if you're if you're especially I would say if you're trying to make like, you know, old history type settings or things or even historical settings in which because one of the things that dolly doesn't let you do is like controversial stuff, but controversial stuff happened in the past, you know, maybe you want to do art about it. Some people you know, they they get inspiration from that kind of thing and they make beautiful art that describes you know, often in an emotionally affecting terms that that stuff that happened



Watson Hartsoe 34:50

that's that's interesting. I like I'm here in Atlanta, and there's like the Martin Luther King national like Park, right and I wouldn't Claire, and I went to a coffee shop afterwards. And I like took a bunch of quotes from Martin Luther King and from Gandhi, and like, some of them wouldn't even let me put these quotes.



35:11

This is yeah, this is a part of the dumb, the dumb word filter approach, which the mid journey claims that they're gonna I don't know what Dali is going to do majorities claims they're going to loosen it. I will believe it when I see it. But yeah, there's a lot of inelegant hodgepodge solutions right now. And so I, I just, I don't know, I don't know, if there is a solution, maybe one will come along. I don't necessarily think it's feasible to just say, okay, but no AI is until we solve the problem of, you know, the problem, because these are neural networks, you know, what they are dealing, what you're dealing with is the problem of overfitting associations at the neurological level. You're so if you say like no AI is until we solve the problems, like, how do you solve the problem of bias at a fake neurological level? There's a lot of social psychologists who would love to know the answer to this question, too. You know? And, again, because the, you can't you just can't keep a lid on this stuff. So I don't think that's feasible. I think so I think you have to be vocal about it. I think you have to be aware that it exists. But I and I think I hope that there's a solution a little bit more elegant than dollies around the corner. Again, I think it has its merits and demerits. But I don't know that there's a knockdown solution yet.



Watson Hartsoe 36:27

I was I was impressed with the idea that you had of like, you know, throwing in this, this random dice roll of like ethnicity



36:37

eventually started doing Yeah, okay. I don't think I posted about this in like a 1000 club or something at that time. But I was I was pitching about Google's, because Google has a model there had two models, actually. They're not like tuned, publicly or anything yet. But they have models that are very powerful. They have bigger parameter sizes than any of the publicly available ones, but they don't want to release them because they're worried about the bias problem. And like, I have my disagreements with what Dalek done, but they tried. And the thing that irritated me was like you guys are, how is this not an obvious kludge solution to you the people computer science just like take the underspecified words, and do a dice roll in it and pass the dice, you know, you will just mean that they didn't specify gender, so 50/50 chance of a man or woman, you know, specifying ethnicity. So whatever the percentages are there, maybe just do one in four, one or three or something I don't know, race is fake and a social construct. So the numbers are arbitrary. But, you know, there's there's just there's things you can do to mitigate that there's steps you can take, and I think it's a little precious to be like, well, we have a super powerful, it's the best, but you can't see it right now. Because Because nurse might prop too many women. And it's like, people know that these lynxium, you know, you're the company that runs the image search, you know, do a Google Image Search for carers right now, what am I going to see? You know, that's this, it's going to be the same thing, you get out any of these things. So yeah, it's, I mean, I think I think there's things you could do, I think, the most important thing is just to be aware of it. And also, and also not to make it worse, because you got the, the, the Hippocratic principle or whatever it is, first, first, do no harm. That's part of the reason that I did like the default stylized decision, because I think if you one thing that definitely isn't going to fix it is opening it up to like, a public vote on you know, what features make a girl beautiful, and incorporate those models so that it's more likely to generate those when you when you trump the word beautiful that I don't think is that's not a good road to go down. I do not like if you make these, you're not going to fix this problem democratically, because the problem that arises democratically.



Watson Hartsoe 38:52

This is a good framing of this situation. I think that I'm reflecting on the idea of like Google Image, Image Search, doing the same thing. But in a different light, the idea of it being generative or not existing or being almost like a dream, I don't know if that makes it more disturbing, or why this is like a place where this is an issue whenever it's not an issue in search,



39:14

maybe shoes that they when they try it, because at one point, I think it was a minor meme online that if you like search for inventors, you would get black inventors. And this turned out to be an artifact of how Google was like doing its because these days Google less and less gives you like straight up searches and tries to like finish things off and give you custom, like views of stuff now. And it was an artifact of how they were doing that. But like, again, there's there's all these unintended consequences, the minute you go down in and start monkeying with the ISTEP. You know, the associations that are that are trained versus the ones that you impose on Facebook, those interact even less predictably sometimes.

 Watson Hartsoe 39:53

Yeah, I remember there being this case where somebody had Google search, what's the ugliest language in the world? And then he can come back with this Indian language I think was Canada or Canada? I don't know how to say it. But apparently, like this region of India took incredible offense at this. And like, I mean, rightfully so. But like my thought is like, the more the more people have access to this, the more likely you're to be able to find extremely offensive things that like people weren't even thinking about before. Yeah, I mean, I don't know, like, I'm curious, like how you think about the idea that this can maybe be used to also like create filter bubbles, or sort of like filter realities, or like, sort of like just surround people buy images of what they want to see in a way where they never have to look outside of that.

 40:41

I think that's definitely a, that's definitely a concern, as well as people. Just want I hadn't thought of actually, now that you mentioned it, that is kind of disturbing, is the idea that, you know, again, I consciously try to have some variety in what I prompt. And I'm sure there are people who will just sit there and generate the same thing 500,000 times and not get well, probably not so much with this because again, I don't I don't necessarily think the problem is still images, because still images are only so unless you're really like a like a big comic fan or something like that, you know, most people live, the killer app for them will probably be in 20 or 30 years, when you can generate, you know, you can have generative Marvel blockbusters, or whatever, you know, the hyperstar, hyper saturated, all stuff that's like saturating your mind and your ears, and maybe the haptic feedback will be good enough that it saturates all your senses, I don't know. You'll get the Oculus, the Oculus 2050 was smellivision.

 Watson Hartsoe 41:45

We'll see we'll see when that happens, what people are what people are prompting as smells have been interesting, I

 41:52

don't know if you can address that concern. Because again, it's just a, it's just an accelerated version of the concern that we've always had, which is, you know, you got, oh, well, my kid or my, or my relative, my dad just like, sits in his room all day and just watches, you know, watches his shows or whatever. We watches the same thing over and over again. And you know, they could they could stand up, get out more or branch out? And you know, I don't? I don't again, I don't know that a lot of us are ready for and prepared to withstand that kind of, you know, why are heading temptation, but I also don't know, what can be done short of talking about it. And, you know, socially reinforcing that, hey, this is not a you know, cool, cool guys do multiple things. Will guys watch multiple things? I don't know if there's any top down solution for that kind of problem. I agree. It's going to be a problem.

 Watson Hartsoe 42:45

I mean, one thing that's been curious that I mean, I've I've felt like the ability that I like, I'm not addicted to my journey. I've certainly heard people mentioned they felt addicted to it, or, like the word comes up in a light way. You know, it's not like a total thing. But I do get a sense. Like, there's a certain, I don't know, like, slot machine feeling to it, like where you're like, wow, like, I'm gonna win, like, I'm gonna, like, get this like novelty or dopamine, or I don't know exactly how that mechanism fits with your idea of Prop craft. And like, I mean, I definitely sense that like, sometimes I like, while I'm amazed at how much time has gone by like, and



43:24

it's a little bit more frustrating than anything else right now. Because one of the things that you have to I mean, I used to it used to be that I thought I was being you know, rigorous Ralph by prompting everything with the same seed. But these days, and especially with the new algorithms, because test and test pier like this, and I'm pretty sure that this is I'm pretty sure that this is a downside, I don't know that it is I'm not a coder, I don't, I'm just a guy who uses a couple of different versions of these things. But I suspect that as the models get bigger, and generate more of their own associations and have more data, that deed noise, which is currently quite determined in mid journey, but much less interpreted on unstable diffusions. And on the mid journeyed stable diffusions, which are tests of tsp. Eventually see noise is going to matter a lot less and so even things that have the same seed are going to be wildly divergent. And these days, I'm not even if I'm testing something, I'm not satisfied until I run it with the same seed four times.



Watson Hartsoe 44:28

Interesting. Yeah, I'm like, like, I'm really impressed with the rigor, like I thought I had been, like, you know,



44:35

grid soon, is that you'll be able to do four by four in one shot, which be much easier.



Watson Hartsoe 44:41

Yeah, I mean, like, I don't know if it's like Citizen Science or how you picture the relationships, like being empirical with this practice. And like your, your ability to, like bring rigor to it.



44:52

I think you have to, I think it has to be explained empirically. I don't think it has to be used empirically, or optimally, or anything like that. because a lot of it is random, a lot of it is, well, a lot of it is both indeterminate. And it's also in a literal sense that it's also just random in terms of the associations that get made. One of the things I was doing today, just messing around was just prompting country flags.



45:16

You know, I have the I have these widescreen



45:19

lyrics prompts that I do, where I just crank chaos up, and I crank the quality out, and I just put it song lyrics. And I get really cool visuals out of that, again, this is a completely This is a completely unscientific way to use mid journey, I arrived at it like, it's kind of like, okay, well, chaos is good at filling in all these details and making things unpredictable and you know, quality will generate epic scenes and give it more time to get some sense out of the lyrics, because it can do that. And it's good at it. And so, one of the things I was doing this morning was just library, just prompting country flags with those same settings, I eventually had to up the chaos a little bit because it wasn't, you know, getting something I was you could do things like you would prop, the Brazilian flag and it would just show you a picture of Carnival, or a picture of a beat a tropical beach or something like that. Or I would probably I probably hit Greece at one point, just a Grecian flag. And it gave me a nice little visual of like, looking out from a I don't know, a hotel or something with like beads in the Windows on a dock.



46:22

It was really cool. So, yeah,



46:25

I think I think you have to be I think it's important to be a little bit rigorous if you're trying to to be a guy who is explaining things and teaching people, but using it, I think it's I think it's totally valid and fun to just do what you feel like, just bounce things off of it and see what happens. That's, that's a huge part of the fun.



**Watson Hartsoe** 46:47

I agree. I'm curious, like in the Gulf of explainability, how people will fill that with like magical or superstitious thinking. And I'm wondering how, yeah, how like you encountered that or how you think about, you know, people whose prompts, like,



47:03

people are stealing them, it's like they they see a guy who does a prompt, and it has all this junk in it. And he got a really good result. And they just take that stuff, if they don't know which one of them mattered and which one of them don't. And you could figure it out, you could like run that same prop with the same seed and take them out one by one. But you know what the token limit as long as it's hard to do. And also some of this stuff comes from the interaction of certain words, or their likes or something is as vague as their place in the list in the laundry list,

you know, because you left to right. So there's just so many variables that you know, I, I made fun of a little bit, but I honest, but honestly couldn't tell you like if someone has a long prop with training on art station and all this other stuff in it. I couldn't I mean, I could give you some hunches based on what I've done, but I couldn't sit there and tell you what percentage they're contributing, you know, I don't know. So they got it, the best you can do is just steal it, and then maybe tweak it and they'll tweak like a subject or something. Or maybe they'll maybe they'll like, eventually, they'll get to a point where they have a really long subject, I'm gonna have to start selectively throwing things out to make room for it in the in the token limit, you know, and it might have to get a little bit empirical. But I think it's perfectly fine to to just take prompts from I mean, I do that too. It's again, it's a thing I do. I just I try not to do it in the manual, just because I'm trying to be like, Okay, well, there's not a lot we know, but there's a couple of things that we do. And here's some really simple toy examples to show you why we think they do or don't in some cases.

#### Watson Hartsoe 48:36

I mean, do you think that? I mean, like, like you said before, like, all this stuff is changing so quickly. And like it varies from algorithm to algorithm, this idea of prom craft being a language that that might be, you know, very, very different dialects or, like it might,

#### 48:53

I don't know that it's be a thing. I mean, right now, you have to think a lot about what you product. But we already know now like it's already this is we're already insane levels, natural language processing compared to how you would have had to command something like this in like the, if the horsepower had been there in the 90s, then, you know, you would have had to command this stuff much more would have been much less natural, you know. And so I think there's, I don't know that this is going to happen, but I think it's impossible that it might happen that just the maybe the first thing to catch up is the natural language processing. And this whole prop crafting thing won't even be a thing in 2025. Like maybe it will literally just be you tell it what you want. And it's good enough now in processing English to know that when you asked for like, I don't know, an orange sweatshirt, the odds are you're not asking for a sweatshirt with an orange on. Because that's just a thing that you know if someone says an orange sweatshirt, that's what you think of thinking of the color. I mean that knowing when you're talking about the color and when you're talking about the fruit is just ultimately a nap neural processing thing that's just, that's I'm, I'm one of the, I think dolly to is probably by far the best at this, right now, you can prompt that those three baskets of fruit with sorted fruit and you can tell it what fruit you want and the three baskets and it'll do it, you have to wait a little bit unnaturally if the bump a couple of phrases near the top, because dolly does have left district left or right word order, and you can't wait things in there. But this shouldn't be too surprising, because they forget own GPT three, of course, they're going to have the best natural language processing, you know. So it's entirely possible that this won't even be a thing and a couple of years. Now, the Office Hours don't talk a lot about because the guys next leap motion guys will talk about like, you know, Leap Motion style interface, I don't know that that's going to happen either. I think it's most likely just going to end up at like very natural, very natural language processing with maybe a G UI element. Once things get to the point where you can refresh the image fast enough on Compute, to like, see what your changes are doing in real time. Once things get to that level of speed, that you might start seeing things branch out from

the text command line interface to, to having, you know, actual brushes that you can point it at or Spacek stuff, you know, you can start out with in painting right now, but it's a clumsy, it's not a real time process. It's a clumsy thing where you download it, you erase it, and then you upload it and then things for a while, and then it fills it in. It's not. You can't operate it like a clone stamp or something in Photoshop, but you will be able to eventually, pretty sure that



Watson Hartsoe 51:32

I mean, that's an interesting idea that, you know, it's just going to be so good at knowing language that for me, like the the sense of like the like poetic coefficient, or like the interpretive flexibility of language becomes like really relevant to like thinking that maybe that might limit creativity or make creativity this more literal



51:53

to but again, the lyrics things like the lyrics prompts, have more or less turned me around to the idea that the creativity is not going to be the hard part, to automate. Creativity, I think he's already there in a, to a great extent, especially with mid journey. And this is why I ultimately why I like it more than the other AIS right now. Because Dolly, I don't know if this is the rails or it might be because they overtrained their model. But it's very, it's hard to get literal concepts match together that don't go together as Progen and stable diffusions. Like these two, they're prone to just ignoring. And so there's maybe this is a function of skinny models like they have here. But it's very good at combining concepts in a novel way, which is things that you associate with creativity, or it's very good at taking an underspecified you know, which again, directors are often prone to do, you know, make it pop or whatever. It's great at taking a vague notion and just like filling in some details that look cool. So I mean, I think the hard stuff is going to be the intelligence, not necessarily the creativity, which is not something I would have said before I started with this.



Watson Hartsoe 53:05

That's curious, I really appreciated the metaphor that you had at the beginning of the the unofficial manual of like the right brain, like slightly stone diffuser, and like the left blank brain like Egghead jazz, like I was like, that's such a good image for making



53:20

a long time ago. And I'm like, Yeah, but it's too good. An analogy of staying in.



53:25

Oh, you said it was just kind of in a while ago.



53:27



53:27

Apparently, apparently the whole thing with like, the left side of the brain does this and the right side of the brain does this apparently this was like an old, like pop science thing that has since been like you're just proven were shown to be way more complicated, which doesn't surprise me. But it's too good analogy. So I just left it in.



Watson Hartsoe 53:44

Yeah, that that definitely clicked in my head. So I mean, like in using that analogy, despite its you know, maybe inconsistencies with science, like the idea of it, mid journey, maybe being more right side than dolly being more left side, I don't know if that's like,



54:00

be it like US dollar US dollar use a similar set training set to majority but they didn't open it, I had a million, a million a million dollars with a GPU resources or whatever to dedicate to this. And so they, they trained the hell out of me when the natural language processing is better. That might also be what makes it a little less creative. No, stable diffusion has a much bigger set, I think it might be a bigger set that you can train as law. And it also has a problem with that. So there might, they might have caught lightning in a bottle here. Like they they're good at keeping old versions of the algorithm around. So I'm hoping that you know, I'm always going to use this, you know, even even if the next one's better, I'm hoping the next one can keep some of the spirit or most of the spirit for sure. But it might just be a function of model size. Honestly, it might just be that you have to there's you know, the big models are good for photo realistic stuff and clipart and things like where you just want the thing that you You said, but if you want creativity, you have to go to a fuzzier Mall. Because you're just not going to get that same kind of random inspiration from a thicker model or a stiffer model, or I don't know what the vaguely phallic metaphor is, but, you know,



55:21

you're just not gonna get it from most



Watson Hartsoe 55:25

interests. I mean, let me let me back up for a second. And like, how did you come to understand or interrogate? Or audit? Or how did you get to the place where you could? I mean, I guess like, your profession helps you with making sense of these systems. But yeah, I'm just blown away with like, the level of understanding that seems like such a black box.



55:47

I cannot code I know people who code and if there's a question where I think my lack of be Well, I mean, I can like do dumb shit. Like I can write, I could write a batch file or something to automate and upscaler. And I took two years of C++ in high school a million years ago. But I'm

not I'm not I don't know Python. I don't know. I've never built one of these neural networks. My understanding is just kind of reading papers and glazing over the math. And then talking to friends of mine who can code and just asking them about it. Like, is my just distance does this analogy sound? Right? Yeah. Well, it break things down. But the analogies fine. So yeah, just just just reading a bunch of stuff, I guess. And hearing them talk about it, they were there were a little bit more in the in the discord has a history, you know, I joined in June, but there's a history of the discord that you can look at and search. And so sometimes that would be like looking for trying to figure out how stuff worked. And I would do a search for it. And I would find it, you know, all in older posts in the database, because they're a little bit more forthcoming about how this stuff worked back then. Now, I think they're a little bit more protective of their stuff, very secret sauces. Because they have more competitors now obvious. Although I'm heartened that they're doing their own thing and should not call it's an understatement. Fusion four. But yeah, a lot of it's just just reading a bunch of stuff, really, it's kind of what I do



Watson Hartsoe 57:18

in the community has been an asset and understanding and explaining, like



57:24

I said, I didn't invent the rolling things and do it prefer option set, I was just doing very specific things with it. There's other things that I probably picked up from the community, like



57:40

what am I thinking of? You overlapping subjects



57:44

thing, that was the thing that clarinet I had bouncing off of each other for a while was What does overlapping stuff in a multi crop like do for you? Does it do anything for you does advantage anything. And he got the notion that you could take a character and Puppet them around, which I think you can basically do, although it's obviously it's limited, and it's not everything, you know, everything and there's no, there's very few surefire solutions to something and mitchard, there's things that will give you that will load your dice, and give you a better shot at what you want. But there's very little, that will just always always work. And then that same overlapping discussion was where I got the idea of like, well, what if you overlap the text or something like that, and then you could put the text somewhere, and you could make a girl holding a credit card with text on it. Then I prompted that and got the text in like two, two variations. And I'm gay. This is fun. And then I immediately posted that and like, yeah, overlapping is cool.



58:39

I'm



58:41

mystified why none of that, as far as I know, like stable diffusion has picked up a lot of tricks. Now, you can do you can do textual inversion and image and which detects the image prompting we have as a weaker version of and train it on characters. There's a front end written by a crazy unfortunate racist who put down his Rimworld whites only mobs long enough to quote a friend for stable the future that has waiting and features like that. As far as I know, there isn't a service that will let you do multiple independent profit targets like Ninja does. That's very fun. And I think it I think, I mean, I don't have any basis for saying this, really, except that I'm a guy who plays with a lot. But I think there's so much more you could do with that, that we haven't figured out yet. Like, there's ways that that interacts with stuff. And all I've done with it right now is dump tax liens and loading the dice a little bit when you want color composition. But I just know that I'm there's, there's, there's, there's that well as deep. There's other cool stuff, you can do that, that people just haven't gotten their heads around yet. And maybe it won't be there until you can use those tools with a better model or something. Because one thing they weren't they were never able to get that feature working with a stable diffusion models. So maybe that's why it hasn't been in any of your Consortium. They probably just want you to think hard to do if they're abandoning them.



Watson Hartsoe 59:57

I'm trying to understand I mean, I don't want to dig too deep into this Because I think I just need more practice. But I think the sense I get is like, it's kind of like layering or specifying where words fit into the image.



1:00:10

Um, it's, it's the way to think of it, it's like this, it's like giving them giving them something they agree on to situate them both in the same thing. It's the like, with text, for instance, you have one side of it, that's just a text. And then you have another side of it is that a text with context. So you have one side of it, that just says K Club, and then that diffuser is only going to work on K Club. And you have another side of it that says K club on the surface of a credit card held by a woman in a department store, blah, blah, blah, in which prompt for the face. So by getting by giving it K club on both sides, I've given both the diffusers, a thing to agree on, and one of them is going to put it in a specific spot, and then the other one is going to pick up on that hopefully, or it's going to put it in this specific spot, maybe the other one's going to follow it, you have no way of knowing which one found at first, because you have to like sit there and look at the diffusion screen by frame, which you can't do. But having separated out into like red and green teams or whatever, but so you give them that agreement. So now the one diffuser that just has the word on it, that's gonna sit there and be happy as a clam, it's just gonna sit there and it's gonna try and render those words, and it's gonna suck at it, but you give it a couple of V rolls, and it'll be good. Meanwhile, the other one now has a, okay, well, the text was on a credit card. So now it can start drawing, you know, a woman holding it, and it can try and draw the department store, blah, blah, blah, blah. And so you you've given them that agreement, that part of that common ground. And now once they have that agreement, you split them off. And they both do other things it's like so it's like the one of the things that I was trying to fix up for someone in Prop craft was, they wanted an orc woman with green skin and white hair. Well, the

problem that you always have, when you're doing stuff in a single practice, the colors will bleed. And so you'll get green hair, you'll get white, you know, clothing, or white skin or whatever. And so, what I hit upon was the idea that okay, well, you prompt him you do multi prompting and what prompts you say and work woman with white hair. And then another prompt, you say and work women with green skins, so that both of the users can agree there's an orc woman there, but one of them will hopefully pay a little bit more attention to the hair than the other one will. And hopefully, the other one will be busy working on the skin. And then, you know, the idea is that when you they both agree on an orc woman, and they kind of go their separate ways on the different colors that you specify. And that work, I think, pretty, you know, I got three out of four, I think on both of the grids that I ran with that, which is pretty good. You know, when I do that, I'm like, Okay, this isn't not luck, this, this probably works to some extent. So, yeah, it's just it's kind of like trying to give them common ground on something. And then for the less important stuff that's either going to conflict with each other, or just doesn't need to be, you know, prompted the same way you specify those differently, and they go their separate ways. I hardly ever do weighting the site like, you know, these days with the with the specifying the numbers after the weights, I most weight by overlapping the words, that's my weighting system, between left to right, and overlapping words, and repeating them, you can pretty much do whatever you want. Like, I appreciate it, the weighting system exists. I think people get, it becomes a crutch for people who use a lot and don't really understand it, like I did, which is why my Daria misses the journey is just giant head on a booth. And kind of understand at that point, that combat booth. I was like, why, why isn't it showing with combat boots, show me the damn combat boots, and I just prompted combat boots by themselves at double weight. And of course, that gave that that specific diffuser way too much, say and so you just get a woman's face protruding off a giant boom. And it was very unfortunate. I didn't use multiproxy again for a week after that, because I thought, Oh, they're useless, they suck. They don't do anything.



Watson Hartsoe 1:03:46

multipronged, I don't even know if I've used the multi prompt, I think I've looked at it, I have not tested it multiple times just



1:03:53

double colon. Okay. A lot of people only use it for waiting. Because that's, that's like some feature of multi prompting is you can take one part and wait it up a little bit. But it's not all it's not like that's not emphasis that will actually break up your prop. And so a lot of people will be like, they'll prompt like a list of eight things. And then the eighth thing won't show up. And they'll put the eighth thing in its own prompt and make it separate. And then all of a sudden, it'll start dominating, you know, the screen because you've broken it off. And it's really has now effectively given it a 50% influence on top of however you weight it, you know, so right off the bat you've given it usually increased its influence. So you don't really have to use the weights most of the time.



Watson Hartsoe 1:04:33

I see I see like in the manual, the multi prompt spellings for dummies. Is that like that's such a question. What did you say? I said I'm looking at the manual and like the multi prompts spelling

for dummies. That's like so good. Like I've never seen text that good. Yeah,



1:04:50

I mean, it's that that's just it's it's fun. It's fun. Did I did you see this tweet or something like there was a whole thing? People were people were actually salty about this because You know, a lot of people just go to the front page with like, portrait, you know, aspect ratio renditions of pretty girls. And at one point, when I figured this out the front page was all text meets, and people got salty about this. These are dumb. Why are these they're just like, the important the funny part. You know, it's not that they're amazing. It's just that they're being done. That's the fun part. You know, we figured out a way to do text. So hang on, let me see the pasted window here.



1:05:28

Oh, nice.



1:05:29

I figured out how to do this. I made Nicolas Cage with a speech bubble saying we're stealing the sham prompts. And the idea is that okay, well, if people see this, it'd be like, how did you do that? Oh, no, steal it off, steal the way to do the text. And so that's what the front page of mid journey look like that day. It was very funny.



Watson Hartsoe 1:05:46

I mean, this is so interesting. I was just talking to a friend of mine who's a type script printer, like uses, like old school printing presses,



1:05:53

it's really cool is if you're doing a lot of words, it's just about all you can do to get the words right. You know, if you're trying to get four or five words, and more than that, you pretty much got to select particularly for whichever one spell it right. You can't afford to waste the rolls on fussing with your your how you how it looks nice like that Nicolas Cage is ugly as shit is terrible. But he is a little screwed up. But it's still legible. And so I was like, Okay, I've done 10 or 12 heroes of this thing. I'm tired of it. Some people do a lot of the roles. God bless them. I love tinkering too much. So like if something hasn't shown up after a while, I, you know, I don't I don't practice what I preach all the time in the manual where I say virals are really cheap and a great way to develop stuff. And you show us when you should. But sometimes I don't do that because I'm impatient. But if you only prompt a couple of words, if you prefer one or two words, and it resolves on those, and you've only done a couple of roles, and you keep the rolling it, it'll start spitting out five designs that are



Watson Hartsoe 1:06:54

interesting. Yeah, I was saying this was one of



1:06:57

them. This one was only two words. And so once it started resolving it started, like coming up with different ways to render the text.



1:07:06

Interesting.



1:07:08

So like this one is even more conspicuous like this one, I prompted a horror font. And now you're getting a bunch of different spins on a horror font, you're getting the ghostly one, where the top has kind of analysis, the top and bottom has kind of an alpha channel and the first one, and then in the bottom left and right, you're getting the classic steady, you know, I don't know what they're called, because I'm not a font designer. But, you know, there's just cool things that can be done font wise, even. Even if the you know, as long as the text is very short, you can't really I tried to prompt the quick brown fox jumps over the lazy dog wants nothing doing.



Watson Hartsoe 1:07:41

Wow, this is changing my entire perception of what's possible. I mean, I think I'm curious, like, where do you see this going? I think you've talked a little bit about that. But I don't know what you're what you're imagining, you know, a few months down the road versus a few years.



1:07:56

A few months down the road. Like my big thing with this, I want to be able to because I am I like drawing comics or I don't, I don't like drawing comics, I can't draw comics. But I would like to draw comics, I have ideas for stories that I want to tell. And I think comics would be a good medium to do that. And and I think that this is like we're just because I've gotten a character in a speech bubble saying something a handful of times, but it's kind of a pain, and you can't really we're kind of at the edge of that possibility right now. But I'm hoping that the next version or the version after that of this will get it to the point where you can just say this character that I have image prompted saying this in a speech bubble with this background, you know, it'll just be a thing you can do. And with the image problem, the character will look semi consistent. And if once you can get it there, you're off to the races. You can do comics, like completely with this thing. And that that'll be that's my big hope that this where this place will be over the I mean, you could already do a comic that people aren't doing comics now. But like they're doing, you know, they're doing a lot of it in post production, which is there's nothing wrong with that, it's probably a lot easier. But I for some reason, just I liked the idea of doing

everything in the AI, I think that's a, it's a cool thing to do. Even if it's not necessarily practical. It's the same way you got people in demo scenes who like you know, program games and music, like shows and stuff and a handful of bytes. You know, it's not the most it's not the, you know, way you would do that professionally. But it's fun and you did it. And it's a fun activity. And I think I'm looking forward to that. That's what I'm looking forward to the most in the short term. Maybe animation, but I think that the first is going to look like Richard Linklater rotoscoping pretty much. That's what it looks like right now.



1:09:45

Let me find I was first figuring



1:09:52

out seeds and like you, I fed myself into this thing and I generated it with the same CA times and you can see what it looks like there. are, you know, there's a potential there to have live frame by frame rejoins and stuff. And another guy has taken this much farther, there's a guy drift Johnson in, who posted an off topic he treated he used, this is one of those things, it's stable, the future is better. And because it has direct image in it, he did every frame of like music video or something. Just like it's really cool effect where it was just like turned it into like this rotoscoping thing with ideas just blinking in and out, you know, at one point, this would be a column of fire, and then it would be like a glowing, you know, orange tower or something. It was cool. I think initially, animation is going to kind of be locked into that really abstract rotoscope style, which some people will like and some people won't, I think it's going to be a while yet before it gets to the point where you can do more than that, because they that style is an artifact of the line work just being inconsistent from render to render, which is just something that can't be helped. And it's probably something going to get worse in the short term. So traditional animation of you know, taking a piece and animating it by hand. I think that's a steal probably a couple years out at least. And then the movies, I don't know, I'm not, I'm not really that into them. And from what I've seen, I've seen some of the, like, 60 pixel demos and like, Yeah, but they've got the kind of the same problem as the animation does, where stuff is just kind of flickering too much. And there's, I don't know how you because because the one of the things that I learned when I was reading the papers for this is that the fundamental resolution of these things is very small. It's like It's like 64 by 64, or something like that. And everything after that is just like Bootstrap and upscale Bootstrap and upscale. There's scalars, scalars, and scalars. And that's fine. If you're producing, that's why the stuff is very indeterminate. And that's fine if you're producing images. But if you're producing video, like movies, you know, or animations, unless it's in this very specific rotoscope style, it's not going to work for you, the computer needs to have the I would need to have some kind of understanding of how physical reality works. That doesn't need to have for still images necessarily, but it would for any kind of animation



Watson Hartsoe 1:12:03

or movies. What's your sense?



1:12:06

Maybe, maybe though, maybe the movie AI looks like crap now is going to be totally Boston in a year.



Watson Hartsoe 1:12:14

I saw this on Twitter, this guy had fed a mid journey image into this other app called Kadeem 3d. But yeah,



1:12:24

like, that's cool. But like if you I just looking at it like this guy that I don't see any, it doesn't look like it looks like it just kind of spray painted that image onto a very simple texture of an armor or something. That wouldn't be the kind of thing like that's fun. That's cool. But that wouldn't be the kind of thing you could do professionally. Or if you wanted to like code a game or something like that. That guy kind of looks like the oppressed PS to character. Agreed. Like for it to properly do this, it would have to like contour out all that steampunk crap, you know, on the armor instead of just spray painting it all as a flat texture. But I think that's very cool. I think that's probably coming out real soon. Like, I think I saw an NVIDIA paper where they were talking about doing meshes and stuff that are generated and not that are generated and not necessarily model from scratch. They're generated based on images. If you could generate them based on images, then that's that's what I'm talking about here. I'm just talking about higher fidelity. So I think that's coming soon.



Watson Hartsoe 1:13:24

That's the paper with the cars in it. Is that the picture of cars?



1:13:30

Trucks and stuff? Yeah, yeah.



Watson Hartsoe 1:13:33

Yeah, that's interesting. I mean, I think the Yeah, might



1:13:36

not be a top down brand Theft Auto games.



Watson Hartsoe 1:13:40

Yeah yeah The video the video game stuff I mean that's what I was so impressed with

Yeah, yeah. The video, the video game stuff. I mean, that's what I was so impressed with talking to God at play, or Joshua was like, yeah, he's a video game guy. And he's like, transforming his creative workflow. So I mean, like, I'm really curious how different creative professionals are uptaking this? And if you haven't talked to many of them, or if you know, many of them are like, I don't know, architects, I feel like you're kind of off to the races with it. I don't know who else you interacted with.



1:14:08

regard. I don't I'm not I don't work in a creative industry. I'm just an attorney who works for who does oil and gas patents in Houston. And this was just a hobby I picked up. I don't actually I'm not involved with this professional creative stuff at all. I'm basically just big, big time doodler. I doodle little world buildings and things like that. Some of them could finally like some of the characters that I had that in my bed, they were in a spiral notebook that I had when I was unable to pay attention in a common law seminar in law school a million years ago. And now I you know, just call them out to feed some of their characteristics in mid journey and randomized a couple talks until something sort of matches my inner my inner vision of them and bam, they're there. They're real people now.



Watson Hartsoe 1:14:58

You think like with expertise, like To the there would be like a path open for you. And like in a creative industry, if you continue with your prompt crafting and knowledge of this,



1:15:08

oh, it would be nice to think there is. But I don't know, like, Hey, I don't know how I don't really know if these things are going to be thing for awhile, Walker, and B. I think that this is not going to replace humans. I think this is going to complement them, I think mid journey and other AIS, they're going to be in the room, they're going to be in the artists room, or they're going to be in like you put GBT, three of the writers room or something like that, you know? I've experimented with that a lot. I think it's I think it's I think it's underrated, I think especially if you crank the I think GPT three is very underrated and creativity, if you are willing to kind of like treat it like, I don't know what what's the Damascus steel or whatever cracked the creativity of so it generates nonsense, but there's got to be some great little bits of reports a advancer in there that you can refine down, you know, when with editing. This is similar to that I was reading an interview with The Simpsons writer who said the way he writes things is he uses just crappy form dialogue and utilitarian prose about that he said this, okay. And then he's done it, he's done it in the crappy way. And then he go and rewrite it, which is a funner job than writing, rewriting is great rewriting is easy. So like, I think the majority is going to be in the room. But as far as a role for me, like, because the analogy here is going to be kind of a creative director. But I think to be a good creative director, you'd have to be able to not just know how MIT journey does things, but also know how the other humans you're directing do things. So you know, me going from a because like, job with director in the title, you know, from just being a schmuck that attorney somewhere, I don't know if that's viable, where I know that anyone wants a 40 year old rookie, on their, on their creativity in this economy. So I think guys to think about and I'd be, you know, open anyone who thinks that of me, but because, you know, I don't like the the penetrating flex, but don't put that in. Good. Yeah, I don't know that it would be I

don't know that it's gonna be a thing for that long. And I don't want to have loads of sorry, my cameras are talking to fly very noisily. failing at it. Yeah, I don't know that that's gonna be a thing. It'd be fun. If it is. And if it sticks around for a while longer even maybe try and do something with it. I think the little story I have to tell about, you know, Pat, attorneys, that'd be a nice thing to talk about. But again, a lot of this, a lot of this stuff is very, like a lot of this stuff is real, you know, is operated really in a lean way. So, you know, maybe the thing that to be doing is just like independent consulting or something, it's like, I don't know that any I know that there's people who are like employed by Google as something is like prompt engineers. And you know, those, those SOPs get to sit all day at Imogen and party and just go ham, I'm sure. But the job listings I've seen for those have mostly been well, okay to be a product engineer, we also want someone who knows how to code, which is fair, because if you know how to code, you're going to know how to explain certain things in ways that I can only just sort of Intuit and guess at which and I think it's going to be like, it's going to be similar to that with with creative places, except more of the director thing, like they're going to want to, you're gonna want to know that, you know, could talk to an artist just as easily as you could talk to a computer, that part is not quite as easy for me.

 **Watson Hartsoe** 1:18:25

I was talking to somebody who works at a world building agency, and they just like build worlds for films and video games and help companies imagine the future.

 1:18:36

And how, huh, missed out? Why did I go to law school,

 **Watson Hartsoe** 1:18:40

you can still I mean, it's still around. It's a, it started as a production design business, the guy who did Minority Report, he got a bunch of scientists and different types of thinkers in a room, like had them like, hey, let's just do some good research and, you know, speculate a little, but I was talking to this guy, he, I think he works at the Salk Institute now, but he was he was in the room when they were doing some, some creative world building project under NDA. So he didn't tell me what it was about. But he was just saying that, um, they they started out everybody on the team was using mid journey. And it was like frenetic, and everybody was having a great time, but they had to actually shut it down. They just shut it down. Because it was like, everybody was just focused on their prompts and like getting lost in proc craft, they weren't talking to

 1:19:26

each other. It's a really good concept artists, I think that's it's bread and butter. Now and probably will be for a while. But you know, at some point, you got to take a concept and say, Okay, this is good enough, or we can refine it to be good enough. And then let's go let's get to the, you know, let's get to the meat. So I'm sure Yeah, I'm sure that would be a thing, you know, because I I completely empathize with that because, you know, I've got as as they say,

on Twitter, I've got ADD, so I actually do, but I definitely know what it's like to just have a bunch of ideas and just be too distracted by them. Do concept art on one of them and then just becomes a start hopping from idea to idea never finish anything. Well, speaking

 Watson Hartsoe 1:20:06

of hopping from idea to idea And finishing things, I appreciate your time and I think you've hopped around a lot of ideas and I don't know what what to finish this interview with other than, you know, expressing gratitude for your time and I don't know if there's anybody you would recommend I talked to any other resources. I need to definitely. Yeah, looking, looking

 1:20:28

out on the discord. He's probably the guy who does the most of this besides me. Um, so yeah, he'd be a guy to talk to if you haven't already. And he might, he's a guy. He's like an official ninja to guide now. So maybe he could hook you up with talking with one of the devs I don't know if anyone's been able to do that. But if anyone could, he probably could.

 Watson Hartsoe 1:20:46

Yeah, I'm gonna I'm gonna research team. I think that one of my one of my colleagues is working on it. They're going to interview clarinet on Monday. I think I also is clearly a man or a woman. I don't know if that matters, but I.

 1:21:01

Okay, okay.

 1:21:02

He's mentioned his wife. And I know, I know lesbians exist. But I'm also, you know, Bayesian statistics.

 Watson Hartsoe 1:21:08

The other dice roll. Okay, cool. clarinets have a unique spot. With just the access to the dev team. I assume there's not many people. There's some

 1:21:17

things you can't really talk about with me anymore. Okay. You know, okay, keep your secret. Like I said, they're here, they're a little bit more secret. And like, I am convinced, and I was so convinced about it, that I wrote it in the guide, that like, maturity doesn't matter. And it directly

initializes images, when it does variation roles, it doesn't use them as image prompts in that fuzzy way that it does when you started with an image wants them. They don't want you to be able to directly initialize an image and upload, which is what you could do on stable diffusion with the image the image for various reasons, and that's fine. But they, they definitely do it directly initialize on the former image, they don't just use it as a fuzzy product. And I know this because I tested it for a while and it only makes sense that way. And B none of the parameters kept working on V rolls. But see, I did a search early on. And before the mid journey, Discord was public back in April, a Dev was saying, Yeah, we do. And it's for upscales variations. And so I told this to clarinet. And he was like, Okay, well, let me go bring that because he was tough thinking was image prompts. He was like, let me go bring that to a dev and come back. And they came back with the dev says, that's not what they do. But he can't tell you what they do. And I'm just doing the Okay. Yeah. Okay. That's good. To be as popular as it was. And you wouldn't have to, like, go through and delete, like old posts to, you know, maybe keep some more secret sauce than you felt comfortable spilling now, but yeah, I'm pretty sure that's. So you might be a little bit limited in what he could say. He's a physical guy who doesn't be afraid of anything.



1:22:41

Yeah,



Watson Hartsoe 1:22:42

I mean, I think the secret sauce thing is definitely something to, you know,



1:22:49

there's some stuff that I've figured out because I'm kind of attentive and anal about this. But you know, I'm not like the like the digital painting, which again, is not that's not something I figured out. She's an example of I'm talking about Joe Penna was the one who figured that one out. I don't know if he's on the discord anymore. So I don't know if I could hook you up with him. But he's also a guy who's done a lot of animation type stuff with this really impressive too. And he was the one that figured out that if by doing the blank, blank sign prompt, or the sign saying X, you don't say what x is, you just look since cut off, you got a digital painting there. So that's how we figured out that the devs are definitely nudging this thing towards digital painting. And I explained in that thing, I understand why they do that. It's a legitimate choice. I think it's a good choice. But it's something that people should know about, you know, because if you're trying to do something that's not a painting like a photo, you weren't gonna want to turn that style down. You're not gonna get a photo otherwise.



1:23:46

Yeah, it



1:23:47

 1:23:17  
also joke. Joke on me.

 **Watson Hartsoe** 1:23:50  
You said Joe Penna though, like if I, if I look on Twitter, maybe

 1:23:55  
you see if he's on Twitter.

 1:23:59  
He might, I mean, maybe that was his app.

 1:24:02  
Joe Penna Mystery Guitar, man he's a YouTuber, I guess. Okay. No, he has he has a Twitter it's Mystery Guitar him

 **Watson Hartsoe** 1:24:11  
Mystery Guitar. Yeah,

 1:24:15  
I don't know that he's still working with majority I think he might be working more with stable the fusion now because like I said, he's done animate, he's done really cool looking animation stuff. When stabled fusion is probably the best one for that right now, because of the direct image in it. It makes it much easier to do that Richard Linklater shit. And yeah, he's done some really cool stuff. So it'd be an undergrad. Hello, my name is Joe Penna. I'm a film

 **Watson Hartsoe** 1:24:35  
director. I also I also shared with you that experimental design world building studio, and I know people want to work there. So you know, I think that honestly, this stuff does evolve. They would want they would want your interest. They would want your, you know, your expertise. Just something to adore to maybe think about in the future.

 1:24:53  
See what they say. Certainly, well,

 Watson Hartsoe 1:24:56

I can't thank you enough.

 1:24:58

So in editing, nothing on the beat Just no one can read it and it said terrible. should have picked like four songs instead of one and had a you know, made it a little bit slower anyway.

 Watson Hartsoe 1:25:09

Well, there's more there's more to explore. And, again, I'll send you the research as we as we like coheres, but there'll be there'll be a rough draft.

 1:25:21

There'll be a what I said,

 Watson Hartsoe 1:25:22

there'll be like a rough draft probably done in the next week or two. And you're working on this isn't a book this is like for this is for an online communities like seminar. Yeah, yeah. I'm assuming in human computer interaction. So yeah, it's just like, I'd be interested. Yeah, I would. I would love to get your feedback on it. Hopefully we'll publish something. But again, yeah, just want to share, share what you said back to you.

 1:25:50

Feed them so. All right. Have a good day.